



Immigration and the macroeconomy: Some new empirical evidence [☆]



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ABSTRACT

We propose a new SVAR identification scheme that enables us to disentangle immigration shocks from other macroeconomic shocks in a sign-restricted model estimated on Norwegian data over the period 1990Q1 – 2014Q2. Notably, immigration is an endogenous variable in the model and can respond to the state of the economy. We find that domestic labor supply shocks and immigration shocks are well identified and are the dominant drivers of immigration dynamics. An exogenous immigration shock lowers unemployment (even among native workers), has a small positive effect on prices and on public finances, no impact on house prices and household credit, and a negative effect on productivity driven by a large decline in capital intensity.

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1. Introduction

During the past decades immigration flows have increased significantly in most advanced economies. This is certainly the case for Norway, where the population share of immigrants has surged from approximately 3.5 percent in 1990 to 14.9 percent in 2016, a share higher than the corresponding number for France (11.8 percent), Germany (13.3 percent) and the United Kingdom (13.3 percent) provided by Eurostat. While a large literature has studied in detail the effects of immigration flows on employment and wages using mostly disaggregate data, the impact of immigration on standard macroeconomic variables has not been investigated systematically. This paper aims at filling this gap. We conduct our

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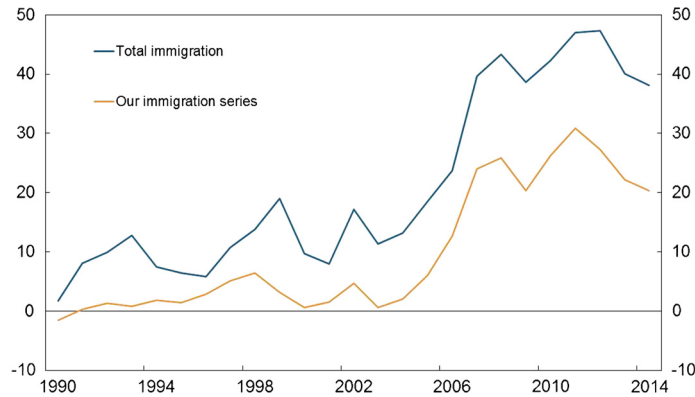


Fig. 1. Net immigration to Norway in thousand people. Our immigration series include EU/EFTA countries, North America, Australia, New Zealand and Eastern Europe.

Source: Statistics Norway.

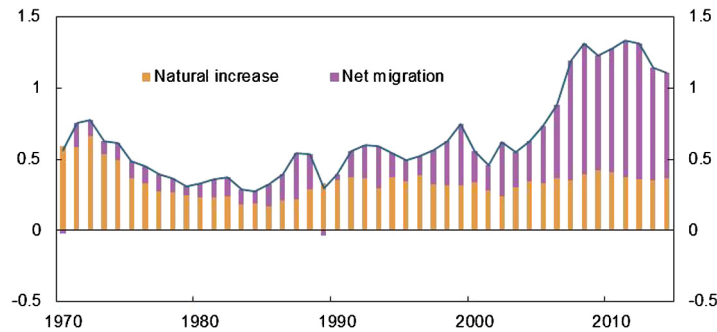


Fig. 2. Annual change in population Norway. Percent.

Source: Statistics Norway.

analysis using Norwegian data as Norway is one of the few countries for which a quarterly net immigration series is available from the early 1990s.

Our goal is to include a net immigration variable into a Structural Vector Autoregression (SVAR) model, which is the most widely used empirical model for macroeconomic analysis. Notably, immigration is a fully endogenous variable in our set-up and responds to exogenous immigration shocks but also to a series of macroeconomic disturbances driving the business cycle. Following Canova and De Nicoló (2002), Faust (1998), Fry and Pagan (2011), Peersman (2005) and Uhlig (2005), our identification strategy is based on imposing a limited number of sign restrictions on macroeconomic variables to disentangle immigration shocks from other sources of business cycle fluctuations. In a second step we complement traditional sign restrictions with a few narrative restrictions, as proposed by Antolin-Diaz and Rubio-Ramirez (2018).

We estimate several versions of our baseline model introducing one alternative unrestricted variable in each experiment. This strategy enables us to investigate the macroeconomic effects of immigration shocks on variables such as unemployment, public finances, house prices, household credit, prices, exchange rates and productivity. Furthermore, we are able to quantify the relative importance of immigration shocks for macroeconomic dynamics and evaluate the strength of the endogenous response of immigration to the other shocks identified in our system. The analysis of the drivers of immigration and the effects of immigration shocks on macroeconomic variables constitute our key contributions.

As already anticipated, our analysis is feasible since Norway is one of the few countries for which a quarterly net immigration series is available from the early 1990s. The series, plotted in Fig. 1, is provided by Statistics Norway. We include in our analysis only the net flow of workers immigrating to Norway from EU/EFTA countries, North America, Australia, New Zealand and Eastern Europe in percent of the population aged 15–74. We exclude from our analysis immigrants from Africa, Asia (including Turkey) and South and Central America since our identification assumptions are most likely violated for immigrants that do not enter rapidly into the labor force (as is the case for asylum seekers, for example). The case of Norway is particularly interesting as immigration was a marginal phenomenon in the 1990s (cf. Fig. 2), whereas it became the dominant driver of population growth in the aftermath of the EU enlargement to include Eastern European countries (cf. Grangård and Nordbø, 2012). In addition, Norway is an interesting laboratory to disentangle the immigration shock from two other labor market shocks. The first is a domestic labor supply shock that turns out to be particularly important, as participation is cyclical and volatile in Norway. The second is a wage bargaining shock that may have a structural interpretation in Norway given the centralized nature of the wage negotiation system (cf. Aukrust, 1977), in

which the wage norm is determined in the sector exposed to international competition and is then used to guide wage increases in the other sectors of the economy.

We disentangle the three labor market shocks (wage bargaining, domestic labor supply and immigration shocks) from business cycle shocks by assuming that they imply a negative co-movement between output and an aggregate measure of real wages which combines wages of native and foreign-born workers. Our assumption finds theoretical support in standard dynamic stochastic general equilibrium (DSGE) macroeconomic models with an explicit role for labor market shocks (cf. Galí et al., 2011, and Foroni et al., 2018) and also in the most used theoretical framework in the immigration literature that features a national labor market with a production function structure which combines workers of different skill and experience levels (cf. Borjas, 2003, and Ottaviano and Peri, 2008 and 2012). To identify the wage bargaining shock we use data on the participation rate following the previous literature. To separate domestic labor supply shocks from immigration shocks we rely on a restriction on the ratio of immigrants over participants which is naturally pro-cyclical in response to an expansionary immigration shock and countercyclical in response to an expansionary domestic labor supply shock.

In terms of impulse responses, several results emerge from our analysis. First, an exogenous increase in immigration lowers the overall unemployment rate and even the unemployment rate for native workers. Second, a positive immigration shock increases public spending in the medium run, but the response of fiscal revenues follows the same path and the net effect on public finances turns out to be even positive in the short run and neutral in the long run. Third, the immigration shock has no effect on house prices, which are mainly driven by business cycle shocks, but also by domestic labor supply shocks that generate a negative conditional correlation between house prices and immigration. Thus, if anything, immigration has had a mitigating effect on the housing boom that Norway has experienced in our sample period. The same result is confirmed when we consider household credit growth. Fourth, an expansionary immigration shock has no effect on domestic prices but results in an increase in the CPI in the medium run through an exchange rate depreciation. Fifth, labor productivity falls in response to an immigration shock. Notably, the decline in productivity is driven by a large fall in capital intensity, thus suggesting that immigration induces the adoption of less capital intensive technologies (cf. Lewis, 2011).

In terms of variance decompositions, our main result is that immigration shocks are non-negligible (although not major) drivers of the Norwegian business cycle, explaining on average around 15–20 percent of output fluctuations. Immigration responds little to the state of the business cycle in Norway, whereas it reacts more to factors that are specific to the Norwegian labor market: when participation by native workers is low (i.e. in response to a negative domestic labor supply shock), immigration increases significantly.

The literature on immigration in the context of macroeconomic models is limited, perhaps due to the absence of reliable quarterly series for net immigration over a sufficiently long period for many countries. Mandelman and Zlate (2012) propose a DSGE model with immigration focusing on the role of remittances for business cycles in Mexico. Earlier contributions include Canova and Ravn (1998), who study the macroeconomic impact of a flow of unskilled migrants in the neo-classical growth model, and Bentolila et al. (2008), who show how immigration flattens the slope of the New Keynesian Phillips curve for prices in Spain. More recent contributions presenting stylized DSGE models with net migration include Bandeira et al. (2018), Braun and Weber (2016), Lozej (2018) and Smith and Thoenissen (2018) with a focus on Greece, Germany, Ireland and New Zealand respectively, while Hauser and Seneca (2018) and Weiske (2017) study the case of the US. In the SVAR literature, Kiguchi and Mountford (2017) provide an analysis on US annual data using the penalty function approach in which a shock to the working population (that could be due to immigration but also to domestic factors) leads to a temporary reduction in GDP and consumption per capita. D'Albis et al. (2015) use monthly data for France over the sample period 1994–2008 in a SVAR identified with a recursive scheme. They find that immigration responds significantly to France's macroeconomic outlook and at the same time immigration itself increases GDP per capita, particularly in the case of family immigration. Two interesting analyses focus on New Zealand, a country featuring detailed data on immigration flows. In the first, McDonald (2013) studies the effect of an immigration shock on house prices in a SVAR identified with a recursive scheme. He shows that an immigration shock has a strong positive effect on house prices and construction activity, thus boosting aggregate demand even more than aggregate supply. The second paper, Armstrong and McDonald (2016), extends the previous set-up to include a second immigration shock associated with fluctuations in Australian unemployment. They find that higher net immigration due to a higher Australian unemployment rate leads to a higher unemployment rate in New Zealand, whereas higher net immigration for other reasons reduces unemployment in New Zealand.

While the literature studying the macroeconomic effects of immigration is still in its infancy, studies using more disaggregate data are numerous (for a survey cf. Card, 2005; Kerr and Kerr, 2011, and Hagelund et al., 2011). Selected issues of interest are the assimilation of immigrants into the host-country labor market in terms of wages and employment, the identification of long-run displacement effects on native workers in terms of wages and employment (cf. Borjas, 2003, and Ottaviano and Peri, 2012, among others), the impact of immigration on public finances (cf. Borjas, 1999; Storesletten, 2000, among others), on house prices (cf. Saiz, 2003, and Sá, 2014, among others), on prices and the composition of demand (cf. Lach, 2007; Cortes, 2008, and Frattini, 2008) and on productivity (cf. Peri, 2012). While we impose as an identification assumption that an immigration boom has a short-run dampening effect on aggregate wages (thus without taking any stand on the long-run-impact on wages for native workers), our set-up can shed light on all the macroeconomic issues listed above in the context of an aggregate time-series approach that is complementary to analysis based on more disaggregate data and with a microeconomic focus. Notably, the main advantage of a SVAR approach based on sign restrictions over alternative approaches is the ability to disentangle the exogenous and the endogenous component of immigration in a system that fully takes into account feed-back effects between different variables.

Table 1
Impact restrictions in the baseline model.

	Business cycle	Wage barg.	Dom. labor supply	Immigration
GDP	+	+	+	+
Real wages	+	–	–	–
Participation rate	+	–	+	+
Immigrants/participants	NA	NA	–	+
Unemployment rate	NA	NA	NA	NA

The paper is structured as follows. Section 2 presents the SVAR model and describes the identification strategy. In Section 3 we propose results for our baseline case with unemployment introduced as an unrestricted variable in the system. Section 4 presents several extensions to discuss the effects of immigration shocks on public finances, house prices, household credit, prices, exchange rates and productivity. In Section 5 we include narrative sign restrictions into our baseline set-up. Finally, Section 6 concludes.

2. The SVAR model and the identification strategy

We start from the standard reduced-form VAR representation:

$$y_t = C_B + \sum_{i=1}^P B_i y_{t-i} + u_t, \quad (1)$$

where y_t is an $N \times 1$ vector containing our N endogenous variables, C_B is an $N \times 1$ vector of constants, B_i for $i = 1, \dots, P$ are $N \times N$ parameter matrices, P represents the number of lags (5 in our specific case), and u_t is the $N \times 1$ reduced form residual with $u_t \sim N(0, \Sigma)$, where Σ is the $N \times N$ variance-covariance matrix.

Given the large number of parameters to be estimated, we use Bayesian methods. Moreover, the model is specified and estimated with variables in levels, as the Bayesian approach can be applied regardless of the presence of non-stationarity (cf. Sims et al., 1990, for more details on this point). We specify diffuse priors so that the information in the likelihood is dominant. These priors lead to a Normal-Wishart posterior with mean and variance parameters corresponding to the OLS estimates. To obtain identification via sign restrictions, we follow the procedure described in Rubio-Ramirez et al. (2010). Additional details about the estimation procedure are provided in Appendix A.1.

In order to map the economically meaningful structural shocks from the reduced form estimated shocks, we need to impose restrictions on the estimated variance-covariance matrix. In detail, the prediction error u_t can be written as a linear combination of structural innovations ϵ_t

$$u_t = A\epsilon_t$$

with $\epsilon_t \sim N(0, I_N)$, where I_N is an $(N \times N)$ identity matrix and where A is a non-singular parameter matrix. The variance-covariance matrix has thus the following structure $\Sigma = AA'$. Our goal is to identify A from the symmetric matrix Σ , and to do that we need some restrictions which are summarized in Table 1. Those restrictions are imposed only on impact, following the recommendation of Canova and Paustian (2011), and are sufficient to disentangle four identified shocks (one general business cycle shock and three shocks originating in the labor markets) and a fifth shock that captures the residual dynamics in the system. All restrictions find theoretical support in a New Keynesian model based on Foroni et al. (2018) that is extended to include immigration shocks modeled as in Kiguchi and Mountford (2017), Łojek (2018) and Weiske (2017). The model is presented in the Online Appendix.

In our baseline estimation exercise we include five series: GDP for mainland Norway (thus excluding petroleum and ocean transport activities, as is standard in all macroeconomic analysis for Norway), the series for real wages (defined as nominal wage costs per hour deflated by core CPI), the participation rate from the labor force survey, the immigration rate series (including only the stock of immigrants living in Norway from EU/EFTA countries, North America, Australia, New Zealand and Eastern Europe in percent of the population aged 15–74) derived from the net immigration series provided by Statistics Norway and the unemployment rate from the Norwegian Labor and Welfare Administration (NAV). A detailed description of the dataset is presented in Appendix A.2. The sample period is 1990 Q1–2014 Q2. We follow the standard practice of using 5 lags for quarterly series, although four lags are sufficient for residuals to behave like white noise. All results are confirmed also when using four lags.

The business cycle shock is defined as a shock that moves output, real wages and the participation rate in the same direction. Its interpretation as a generic business cycle shock relies on the fact that both real wages and the participation rate are pro-cyclical in Norway. It is supposed to capture shocks originating in the oil sector (cf. Bjørnland and Thorsrud, 2016), foreign shocks, different kinds of demand shocks (such as government spending shocks and monetary policy shocks)

Table 2

Immigration statistics Norway 1990–2015.

	All immigrants	Included	Excluded	Natives
Work main reason	34	54	6	NA
Refugees	19	8	34	NA
Family reunification	36	30	46	NA
Other reasons (e.g. education)	11	8	14	NA
Employment rate (2014)	63.4	73.0	54.5	69.4
Unemployment rate (2014)	7.1	5.8	9.1	1.5
Share of population (1990)	3.6	1.9	1.6	96.4
Share of population (2014)	12.4	6.0	6.4	87.6

Note: Included are immigrants from EU/EFTA countries, North America, Australia, New Zealand and Eastern Europe, while excluded are immigrants from Africa, Asia (including Turkey) and South and Central America. All numbers in this table are in percent.

and to some extent also technology shocks (although the response of participation to technology shocks is model-dependent, cf. Christiano et al., 2015).¹

The three remaining shocks originate in labor markets: a wage bargaining shock and two labor supply shocks, one domestic and one driven by exogenous immigration flows.

The wage bargaining shock is defined as a shock that generates countercyclical dynamics in real wages and in the participation rate. These restrictions find theoretical support in a New Keynesian model with search and matching frictions and endogenous labor force participation, as discussed in Foroni et al. (2018), but also in the model of unemployment proposed by Galí et al. (2011). Notice that, as already mentioned in the Introduction, Norway is the ideal laboratory to study wage bargaining shocks given its highly centralized system of wage negotiation. While this shock does not play an important role in our analysis, it allows us to introduce the participation rate into the system in a meaningful way and it guarantees that the remaining labor market shocks are not contaminated by shocks originating in the wage negotiation process (which may also capture fluctuations in unemployment benefits and in matching efficiency).

Our definition of an immigration shock is rather simple and intuitive: it is a shock that generates a positive co-movement between output and participation and a negative co-movement between output and real wages.² Examples of possible immigration shocks are given by the EU enlargement process to Eastern European countries or by shocks originating in key countries of origin for immigrants, like Poland, Lithuania or Sweden, with no direct effect on Norway.

The first assumption implies that ideally we would like to concentrate only on immigrants that enter rapidly into the labor force, irrespective of their geographical origin. However, as the quarterly immigration series from Statistics Norway is only available by country-groups, we cannot isolate job-related immigration. Nevertheless, by focusing on immigration from western countries we capture a substantial share of the recent immigration boom, which is mostly job-related (cf. Fig. 1). Notably, additional *annual* data from Statistics Norway confirm that work is the major motivation for immigration from the countries included in our analysis, whereas this is not the case for the countries excluded from our analysis, as it can be seen in Table 2. Moreover, data from 2014 confirm that the immigrants included in our series have on average a higher employment and unemployment rate than natives, thus suggesting a higher participation rate.³ Therefore, these more disaggregate data validate the identification assumption that a positive immigration shock increases the aggregate participation rate on impact. While a decline in participation from natives is conceivable in the medium-run, it seems unlikely to be quantitatively important as an impact effect.

Our second identification assumption is that an exogenous increase in immigration has a negative effect on real wages on impact. This is in keeping with DSGE models featuring an explicit role for labor supply shocks (cf. Galí et al., 2011, and Foroni et al., 2018) but also with state-of-the-art models of immigration which also treat immigration as a labor supply shock (cf. Borjas, 2003, and Ottaviano and Peri, 2008 and 2012). These models consider a national labor market in which the labor input combines aggregates of labor supplied by workers with different education levels. The aggregates, in turn, combine workers with different experience levels who can be native or foreign born with an imperfect degree of substitutability in production. Models in this literature are built around the assumption that aggregate wages are invariant to immigration shocks in the long run once the adjustment of capital is completed. Nevertheless, these same models imply that the effect on aggregate wages is negative in the short run (i.e. when the adjustment of capital is incomplete) and more so on impact when capital is predetermined, as explained in detail in Ottaviano and Peri (2008 and 2012). In the former paper in particular, the authors present estimates of the effects of immigration shocks on aggregate wages under alternative parameterizations and find evidence of a negative short-run response of wages in all cases. Note that a negative *short-run*

¹ Since all the other shocks imply a negative co-movement between output and wages, it seems sensible to include a catch-all shock which is able to generate the (unconditional) pro-cyclical behavior of wages and participation. The inclusion of other variables into the system would sharpen the interpretation of this shock but at the same time would move us away from the main interest of the paper which is the analysis of labor market shocks.

² Note that we use the mainland real GDP series (and not the per-capita series) in the estimation. It would not be obvious to assume that a positive immigration shock increases GDP per capita.

³ In contrast, immigrants from non-western countries exhibit an employment rate substantially lower than natives and access the country mainly via family reunification or as asylum seekers. Therefore, those immigrants may enter the labor force only with a long delay, in contrast with our identification assumption for immigration shocks.

effect on *aggregate* wages does not necessarily translate into a negative *long-run* effect on wages for *native* workers. The presence of a long-run wage displacement effect on native workers is at the heart of a heated debate in the immigration literature.⁴ However, it is crucial to emphasize here that our identification assumption is imposed only *on impact* and over a measure of *aggregate* wages. We thus do not take any stand on the long-run effect on native wages and, in turn, on the degree of complementarity between native and foreign-born workers.

Notably, the available empirical evidence for Norway finds support for a negative direct partial effect (thus not considering the adjustment of the other factors of production) of immigration on wages. Bratsberg and Raaum (2012) investigate the evolution of wages in the Norwegian construction sector over the period 1998–2005, a period of rising immigrant employment in the sector. They find that a 10% increase in immigration leads to a decline in wages for native workers by 0.6% for given values of output and native employment. Bratsberg et al. (2014) extend the previous analysis over a longer sample period and consider other sectors. They confirm negative direct partial wage effects with a larger impact on previous immigrant wages than on native wages.

We complete the characterization of an immigration shock by including the immigration rate as a separate variable into the system and imposing a restriction on the ratio of immigrants over participants to disentangle immigration shocks from domestic labor supply shocks. Somewhat intuitively, we assume that on impact (and only on impact) an expansionary domestic labor supply shock lowers the ratio of immigrants over participants, whereas an exogenous increase in immigration increases the same ratio. Note that we are just assuming that the relative impact of an immigration shock is larger over the pool of immigrants than over the pool of participants, thus ruling out extreme shifts in the participation pattern of native citizens on impact. Along the same lines, the response of immigration to an expansionary domestic labor supply shock is allowed to be positive or negative. If positive, however, we impose that the immigration response has to be lower than the participation response.

Finally, we include a fifth shock to match the number of shocks with the number of observables. This is a residual shock that does not satisfy the restrictions imposed on the other four identified shocks. Although it is supposed to capture only the residual dynamics in the system, the shock has an economic interpretation: it is a shock that moves output and participation in different directions and output and wages in the same direction.⁵ In this way, the system is fully identified.

3. Results

In this section we present results for our estimated baseline SVAR model and for a series of alternative specifications.

3.1. Baseline model

We plot in Fig. 3 the impulse responses to an immigration shock in our estimated baseline SVAR model. The responses of all variables are expressed in percent, except for the unemployment rate that is expressed in percentage points. An expansionary immigration shock has persistent effects on GDP, real wages, the participation rate and the immigration rate, despite only restricting the impact response for all these variables. The maximum effect on GDP is achieved well before the peak in the immigration response. Notably, the unemployment rate declines on impact and more so after few quarters.

A decrease in the unemployment rate might at first sight seem surprising. However, it is consistent with previous estimates for the US based on a production function approach (cf. Peri, 2012) and results for New Zealand and France in SVAR models identified recursively (cf. Armstrong and McDonald, 2016, and D'Albis et al., 2018).⁶ One may think that a decline in the unemployment rate just reflects the mechanical response due to an increase in participation. However, this effect is small in our model and the response of the unemployment rate is mainly driven by unemployment. The size of the response is remarkable and may highlight some complementarities between immigrants and natives. In addition, another mechanism can help explain this result. In fact, an expansionary effect on unemployment is not necessarily surprising if we consider that many immigrants (in particular from Eastern Europe) are likely to move to Norway with a job offer and enter the labor market as employed. This mechanism is supported by the New Keynesian model presented in the Online Appendix where unemployment decreases under most parameterizations when immigrants enter the labor market as em-

⁴ A continuing controversy on the magnitude of these wage displacement effects on native workers (and on native unskilled workers in particular) is evident in the literature. Borjas (2003) finds large negative effects on native unskilled workers while Ottaviano and Peri (2008 and 2012) find positive effects on all native workers, including the less skilled, due to a limited degree of substitutability between native and foreign-born workers. Note that a zero long-run effect on aggregate wages and a positive long-run effect on native wages can coexist only in presence of sizeable negative effects of immigration on wages of previous immigrants. Note also that the controversy is only on the long-run effects on native wages and not on the short-run effects which are negative (and even for native workers) also in Ottaviano and Peri (2008).

⁵ These dynamics are generated by technology shocks in the models by Foroni et al. (2018) and Campolmi and Gnocchi (2016). However, participation is pro-cyclical (conditional on technology shock) in the model by Christiano et al. (2015). Given these contrasting results, we refrain from a structural interpretation of the residual shock. Nonetheless, its presence in the system is useful, as it allows us to leave the fifth variable in the system completely unrestricted, arguably an important benefit of our specification.

⁶ We estimated two alternative versions of our model with oil prices and oil investments included as unrestricted variables to be sure that our immigration shock is not associated with oil booms. Oil prices do not respond at all to the shock. Oil investments do not react on impact and exhibit a small increase in the medium run following the expansion in economic activity. We conclude that the restriction on real wages is sufficient to set apart oil shocks (which are a component of our business cycle shock) and immigration shocks.

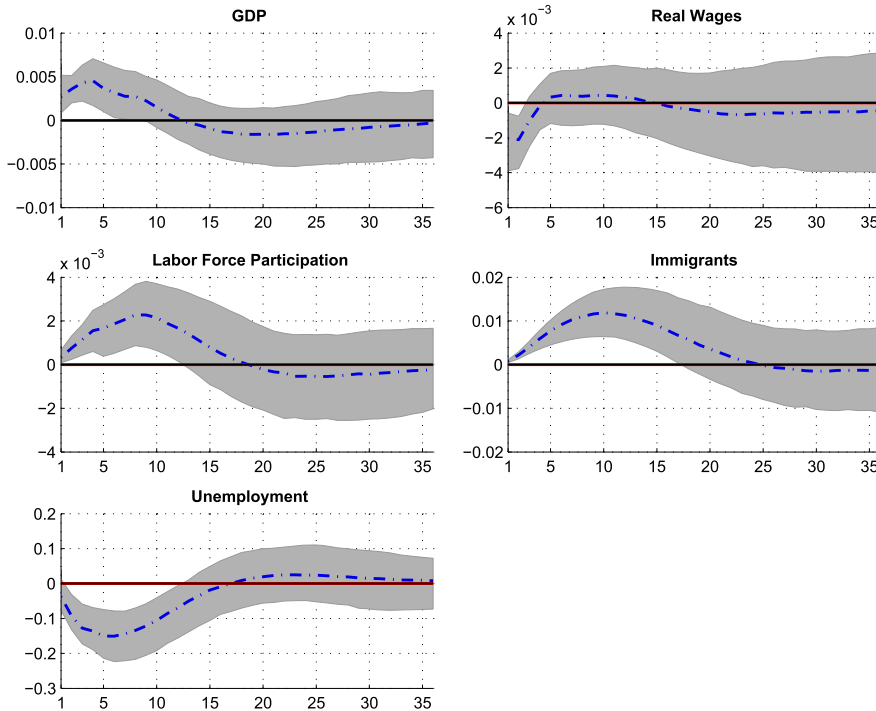


Fig. 3. Impulse responses to an one-standard-deviation immigration shock in the baseline model. The dashed-dotted line represents the posterior median at each horizon and the shaded area indicates the 68th posterior probability region of the estimated impulse responses.

ployed. If they instead enter as searchers, unemployment increases on impact.⁷ Recent data from Statistics Norway also confirm the fast integration of Western immigrants into the Norwegian labor market. In fact, more than 70 percent of new Western immigrants are employed one year after their arrival. Note that the employment rate is approximately the same for immigrants who moved to Norway less than one year ago and for the ones who have been in Norway for several years. All in all, it is certainly possible that the decline in unemployment may reflect complementarities between immigrants and natives but, in addition, it could also capture that immigrants enter the Norwegian labor market as employed.

Let us now try to interpret the other impulse responses. The hump-shaped response in the immigration rate may be explained by the combination of three factors. First, it may be due to registration delays. Registration is in fact mandatory only for contracts longer than six months and it is possible that many workers start working in Norway on short-term contracts (thus contributing to GDP) and only register once they obtain a longer contract. Second, the peak after ten quarters may reflect family reunifications. As we have seen in Table 2, while work is the main reason why immigrants included in our sample come to Norway, the importance of family reunifications is far from negligible. It is reasonable to think that a substantial share of these additional immigrants will not enter the labor force, thus explaining why the immigration rate's response is more persistent than the participation rate's response. Finally, the hump-shaped response of immigration may also be explained by network effects, i.e. the fact that immigrants from the same country tend to follow each other. The establishment of large communities of immigrants from Sweden, Poland and Lithuania may be related to these network effects, in addition of course to the role played by geographical proximity. Moreover, the fact that the GDP response peaks after 4 quarters, well before the peak in immigration, may reflect composition effects due to a decline in labor productivity in the economy that may reconcile substantial and persistent effects on the labor market with a short-lived effect on GDP. We will explore this conjecture further in the next section.

In Fig. 4 we plot impulse responses to an expansionary domestic labor supply shock. The dynamics generated by this shock are similar to those generated by an immigration shock (although substantially more persistent). The important difference is that the immigration rate declines (rather than increasing) in response to a positive shock over a long horizon. The different response of the immigration rate highlights how our identification scheme is successful at disentangling these two labor market shocks.

The immigration rate responds significantly to the domestic labor supply shock but is substantially less responsive to the other three shocks (cf. Fig. 5). It increases following a reduction in the bargaining power of workers and an expansionary residual shock, whereas it declines in response to a business cycle shock. However, all these effects are not statistically

⁷ A similar exercise but in a more complex model is presented in Kiguchi and Mountford (2017).

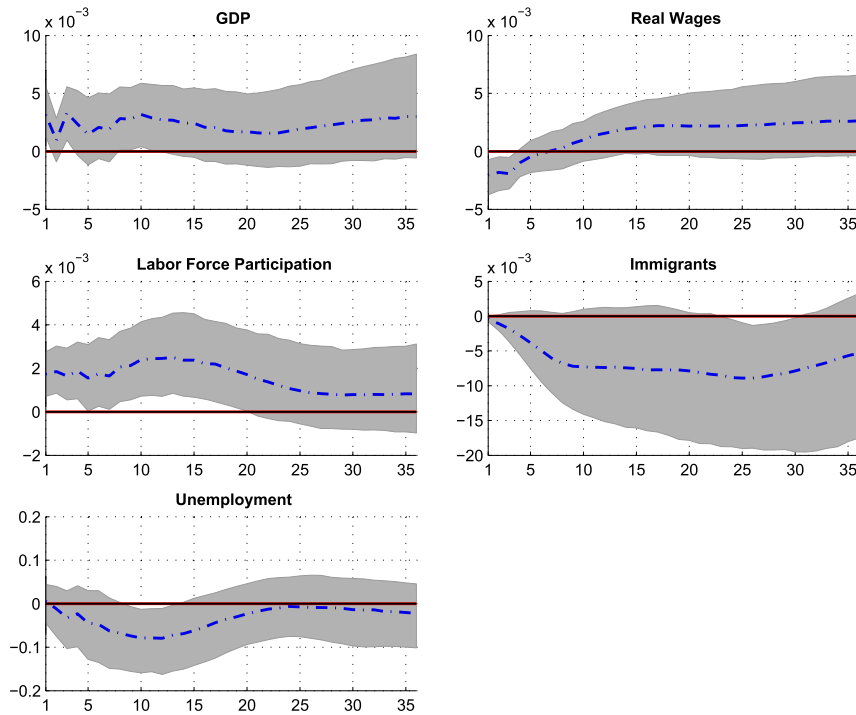


Fig. 4. Impulse responses to an one-standard-deviation domestic labor supply shock in the baseline model. The dashed-dotted line represents the posterior median at each horizon and the shaded area indicates the 68th posterior probability region of the estimated impulse responses.

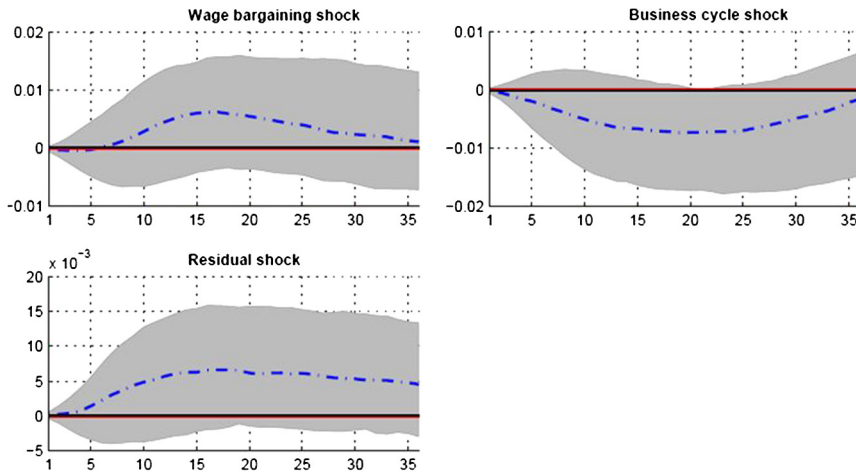


Fig. 5. Impulse responses of immigration to a one-standard-deviation wage bargaining, business cycle and residual shock in the baseline model. The dashed-dotted line represents the posterior median at each horizon and the shaded area indicates the 68th posterior probability region of the estimated impulse responses.

significant.⁸ Notably, immigration exhibits a hump-shaped response to all five identified shocks, thus extending our previous discussion on delayed registration, family reunifications and network effects also to the other shocks.

These results indicate that immigration responds little to the state of the business cycle in Norway (and to shocks in the wage negotiation process) whereas it reacts substantially to domestic labor supply shocks: when participation from natives is low, immigration increases significantly. The muted response of immigration to business cycle shocks is somewhat

⁸ In the Online Appendix we present all the remaining impulse responses. The wage bargaining shock is set apart from the domestic labor supply shock on the basis of the participation response, which is negative on impact and essentially flat afterwards. A decline in the bargaining power of workers lowers unemployment in keeping with the predictions of standard New Keynesian models. In the Online Appendix we also further disentangle the business cycle shock into two components: a productivity shock that moves output and prices in opposite directions and a demand shock that moves output and prices in the same direction. Our main results are confirmed in that extended set-up for which the estimation is more computationally intensive.

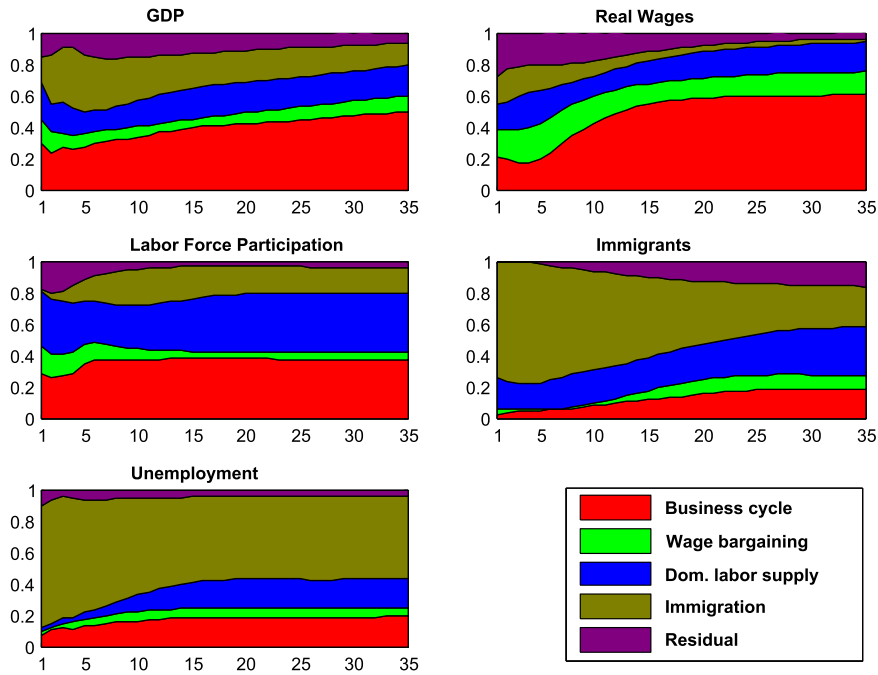


Fig. 6. Median forecast error variance decomposition at each horizon in the baseline model.

puzzling, but may reflect the flexibility of the native labor force. In fact, the marginal participants in the native labor force, who are likely to be close substitutes for labor immigrants, enter the labor market in response to a positive business cycle shock, thus possibly reducing the demand for immigrant workers. The cyclical nature of native participation is enhanced by the fact that schooling is counter-cyclical in Norway.

The importance of labor supply factors for immigration dynamics emerges also from Fig. 6 where we plot the variance decomposition across different horizons as derived from our model. The immigration rate is driven mainly by immigration shocks and domestic labor supply shocks. The immigration shock is also a non-negligible (although not major) driver of GDP and real wages, whereas it is even the main driver of the unemployment rate. The three labor market shocks grouped together explain on average 50% of GDP fluctuations and account for a dominant share of unemployment fluctuations. Such an important role for these shocks is admittedly large but not necessarily surprising given the large number of labor market variables used as observables in the estimation.⁹

The variance decompositions for output and the labor force participation rate turn out to be rather similar, reflecting the large pro-cyclicality of participation in Norway. In contrast, the different profiles of the variance decompositions for output and unemployment (and the limited role of business cycle shocks for unemployment fluctuations) are perhaps more surprising. Those results are, however, consistent with estimates of Okun's law for Norway indicating that a decline in GDP relative to trend of 1% results in an increase in unemployment relative to trend of only 0.25 percentage points (cf. Ball et al., 2013). While unemployment is less cyclical in Norway than in other countries, our results show that the unemployment response is relatively large (and the output response is relatively limited) *conditional* on immigration shocks. This may relate to the negative response of labor productivity, which we will discuss in the next section.

3.2. Alternative specifications

A decline in unemployment in response to an exogenous increase in immigration seems to be in contrast with important displacement effects on natives. Nevertheless, it is interesting to re-estimate our model using a measure of unemployment for native Norwegians that has been available since 1995:Q1 rather than the total unemployment rate. Notwithstanding the difference in the sample period, we remark in Fig. 7 that the decline in unemployment for natives is almost identical to the decline in total unemployment in our baseline model. All in all, we do not find any evidence in favor of displacement effects.

⁹ An important role for labor supply factors in VAR models estimated on US data has been found by Shapiro and Watson (1988), Chang and Schorfheide (2003) and Foroni et al. (2018). Those shocks are supposed to capture mainly demographic factors, but alternative interpretations have been discussed recently in the literature (cf. Barnichon and Figura, 2015).

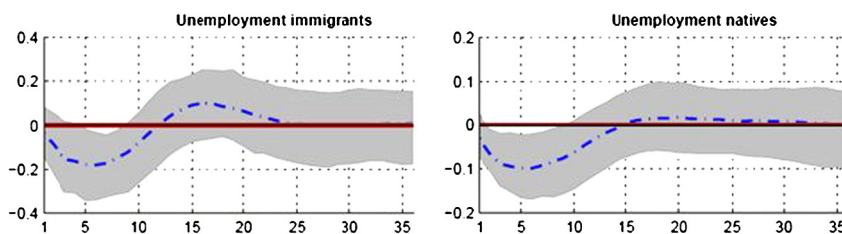


Fig. 7. Impulse responses of different unemployment measures to a one-standard-deviation immigration shock where total unemployment in the baseline model is replaced by unemployment among natives and immigrants respectively. The dashed-dotted line represents the posterior median at each horizon and the shaded area indicates the 68th posterior probability region of the estimated impulse responses.

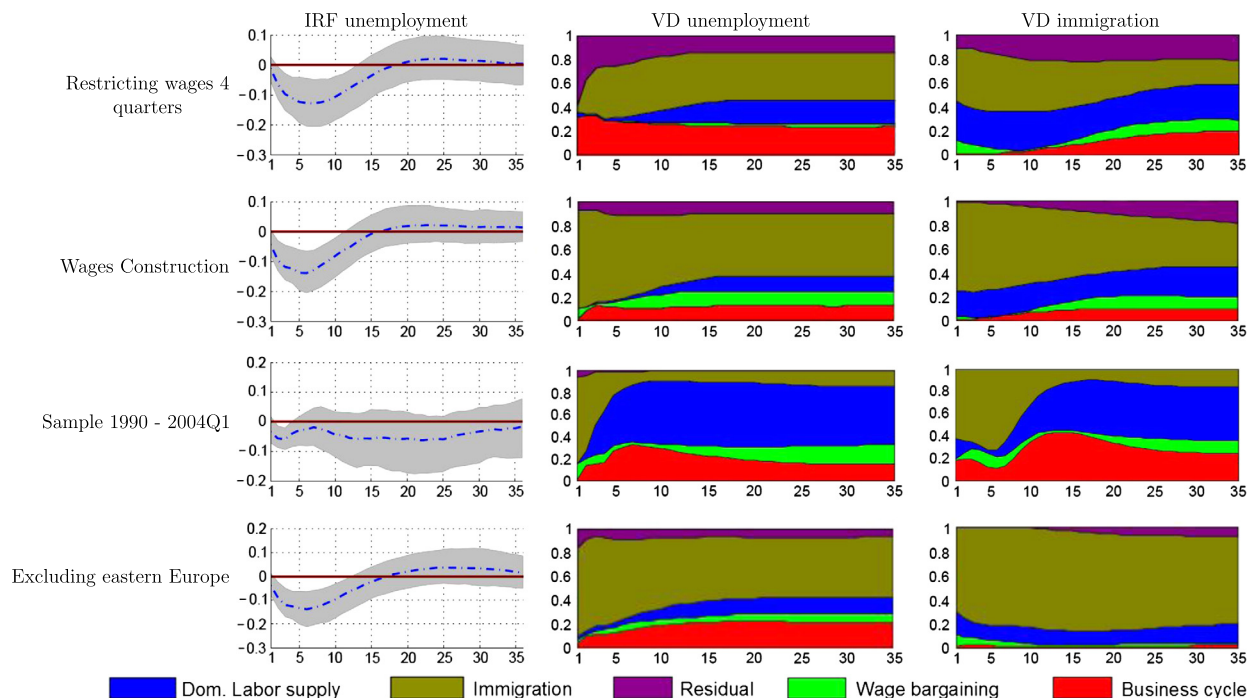


Fig. 8. Robustness checks for the impulse response of unemployment to an immigration shock and for the variance decomposition of unemployment and immigration.

We now focus on our main result by proposing an extensive sensitivity analysis on unemployment dynamics. For each case we plot in Fig. 8 the response of the unemployment rate to a positive immigration shock and the variance decompositions for the unemployment rate and the immigration rate.

In a first experiment, we reconsider the horizon at which we impose the sign restrictions. In our baseline, model we impose restrictions only on impact. This assumption may be more problematic for variables featuring a high degree of stickiness, as may be the case for wages. Therefore, we check whether our main results are confirmed in a version of the model where all restrictions on wages are imposed at horizon four (and only at horizon four). The results emerging from the baseline model are broadly confirmed in this alternative set-up, although the importance of immigration shocks for unemployment fluctuations is somewhat reduced.

In a second experiment, we reconsider our identification assumption on aggregate wages. While fully consistent with models both in the macroeconomic and in the immigration literature, we may conceive a positive effect of immigration on aggregate wages in presence of sizeable productivity impacts (perhaps due to improved efficiency, choice of better technologies or scale externalities) which are usually ignored even in the immigration literature (cf. Ottaviano and Peri, 2008 and 2012). While these effects are likely to play a role only in the long run, let us assume, just for the sake of the argument, that they may be sizeable even on impact. Our model can be adapted even to such a scenario by imposing the less restrictive assumption of a negative response of wages only in the construction sector, i.e. the sector in which native

Table 3

Impact restrictions in the model without data on GDP.

	Business cycle	Wage barg.	Dom. labor supply	Immigration
Real wages	+	–	–	–
Participation rate	+	–	+	+
Immigrants/participants	NA	NA	–	+
Unemployment rate	–	–	NA	NA

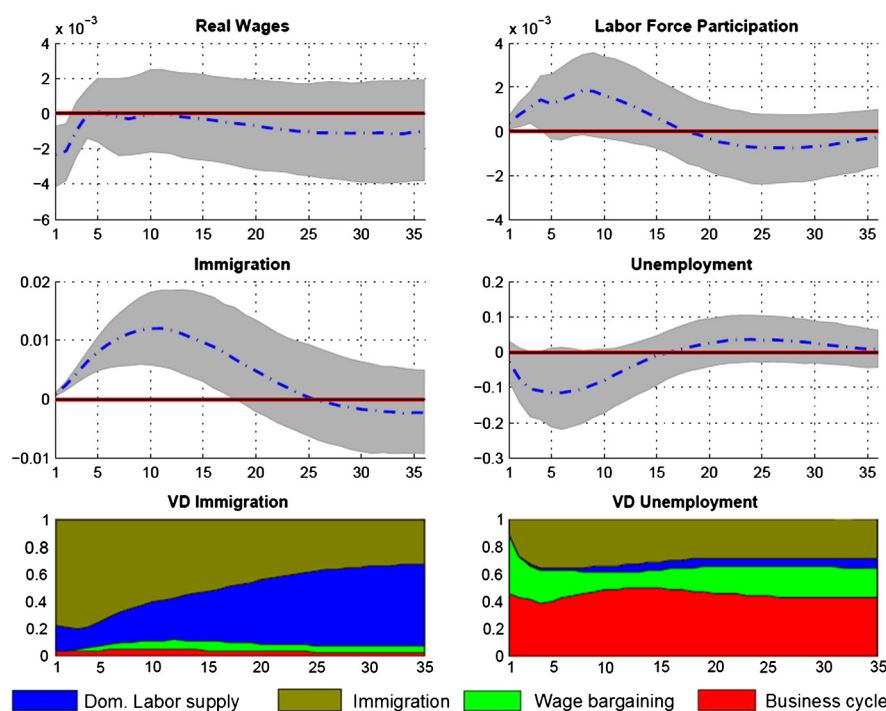


Fig. 9. Impulse responses to an one-standard-deviation immigration shock and variance decomposition of unemployment and immigration in a smaller version of the benchmark model in which data on GDP are not used. The dashed-dotted line represents the posterior median at each horizon and the shaded area indicates the 68th posterior probability region of the estimated impulse responses.

workers face the strongest competition from immigrants in Norway, in keeping with the micro-evidence of Bratsberg and Raaum (2012). Our results are basically unaffected, as shown in the second line of Fig. 8.¹⁰

In a third experiment we consider a shorter sample period ending in 2004Q1, just before the EU enlargement to include Eastern European countries. The response of unemployment is more muted in this case and domestic labor supply shocks are the main drivers of unemployment. It is reassuring that immigration shocks are less important in this shorter sample period, as we expect a relevant role for immigration shocks precisely in the early phase of the EU enlargement.

In a fourth experiment, we exclude immigrants from Eastern Europe from the immigration series. The response of unemployment is unchanged in this case. Nevertheless, we note one important difference from our baseline model. The immigration variable is now driven almost exclusively by immigration shocks or, put differently, the endogenous response of immigration to domestic labor supply shocks is now minor. This hints at the possibility that the endogenous component of immigration may be driven mainly by Eastern European workers who react flexibly to the state of the labor market in Norway.

In a final experiment, we investigate whether the decline in unemployment in response to a positive immigration shock is preserved when data on GDP are not used in the estimation. In fact, one may think that the negative unconditional correlation between GDP and unemployment may induce a negative conditional correlation in response to each shock, including immigration. It turns out that our model can be identified even when data on GDP are not used, as it is shown in Table 3.

Note that unemployment is still unrestricted in response to both labor supply and immigration shocks while it is now restricted in response to business cycle and wage bargaining shocks. The use of these additional restrictions on unemployment

¹⁰ All results presented in the paper (including the ones on house prices, public finances and productivity) are confirmed when we impose the identification assumption on wages in the construction sector rather than on aggregate wages.

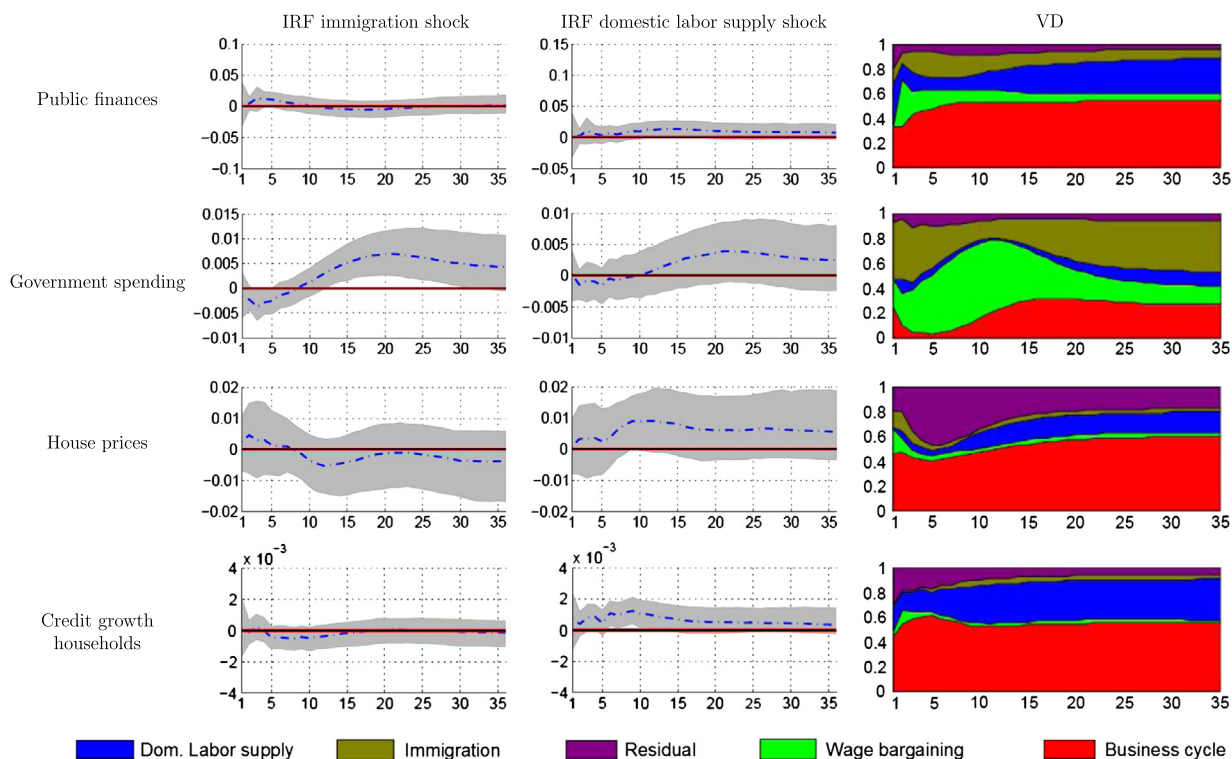


Fig. 10. Selected impulse responses and variance decomposition for some variables of interest.

make the residual shock redundant and identification can be achieved with just four variables and four shocks. In Fig. 9 we plot impulse responses to an expansionary immigration shock and we notice that they are very similar to the ones obtained in our benchmark model. Unemployment still decreases significantly in response to an exogenous increase in immigration, although now the immigration shock is not anymore the main driver of unemployment dynamics. In fact, the innocuous restrictions that unemployment should decrease in response to a positive business cycle shock favors a more important role for this shock in the variance decomposition (cf. Fig. 9). All in all, we conclude that the negative conditional correlation between immigration and unemployment is not artificially driven by the negative unconditional correlation between output and unemployment.

4. The impact of immigration on key macroeconomic variables

In the previous section, we studied the drivers of the immigration rate and the effect of an immigration shock with a special focus on unemployment to discuss potential employment displacement effects through the lenses of our empirical model. In this section, we offer a macro-perspective on some topics that have emerged in the more recent literature on immigration and that have so far been analyzed primarily in the context of microeconomic studies. In particular, we investigate the link between immigration and public finances, house prices, credit to households, prices and exchange rates, and productivity. In each experiment, we include a different unrestricted variable as the last variable in the system at the place of unemployment. The general results emphasized in the previous section are confirmed in all these alternative experiments. Therefore, the goal of the section is to discuss only the responses of the variables that are left unrestricted in the system. The results are shown in Figs. 10 and 11, where in each line we present the impulse response of the variable of interest to a positive immigration shock and to a positive domestic labor supply shock together with the variance decomposition for the variable itself.

Immigration and public finances. The burden that immigrants may place on public finances is often one of the popular arguments used to oppose immigration. Social security programs in host countries (and in Norway in particular) are more generous than in immigrants' originating countries. Borjas (1999) discusses the welfare magnet effects that may attract a large pool of immigrants to countries with generous welfare systems. On the other hand, immigrants may also be net contributors to public finances, especially if young and highly educated. Storesletten (2000) finds that a reform of immigration policies alone could resolve the fiscal problems associated with the aging of the baby boom generation in the US.

We can investigate the impact of an immigration shock on public finances by including a measure of net fiscal revenues (defined as the difference between tax revenues and public spending) as an unrestricted variable in our system. The response of public finances to a positive immigration shock is hardly significant at all and, if anything, on the positive side on

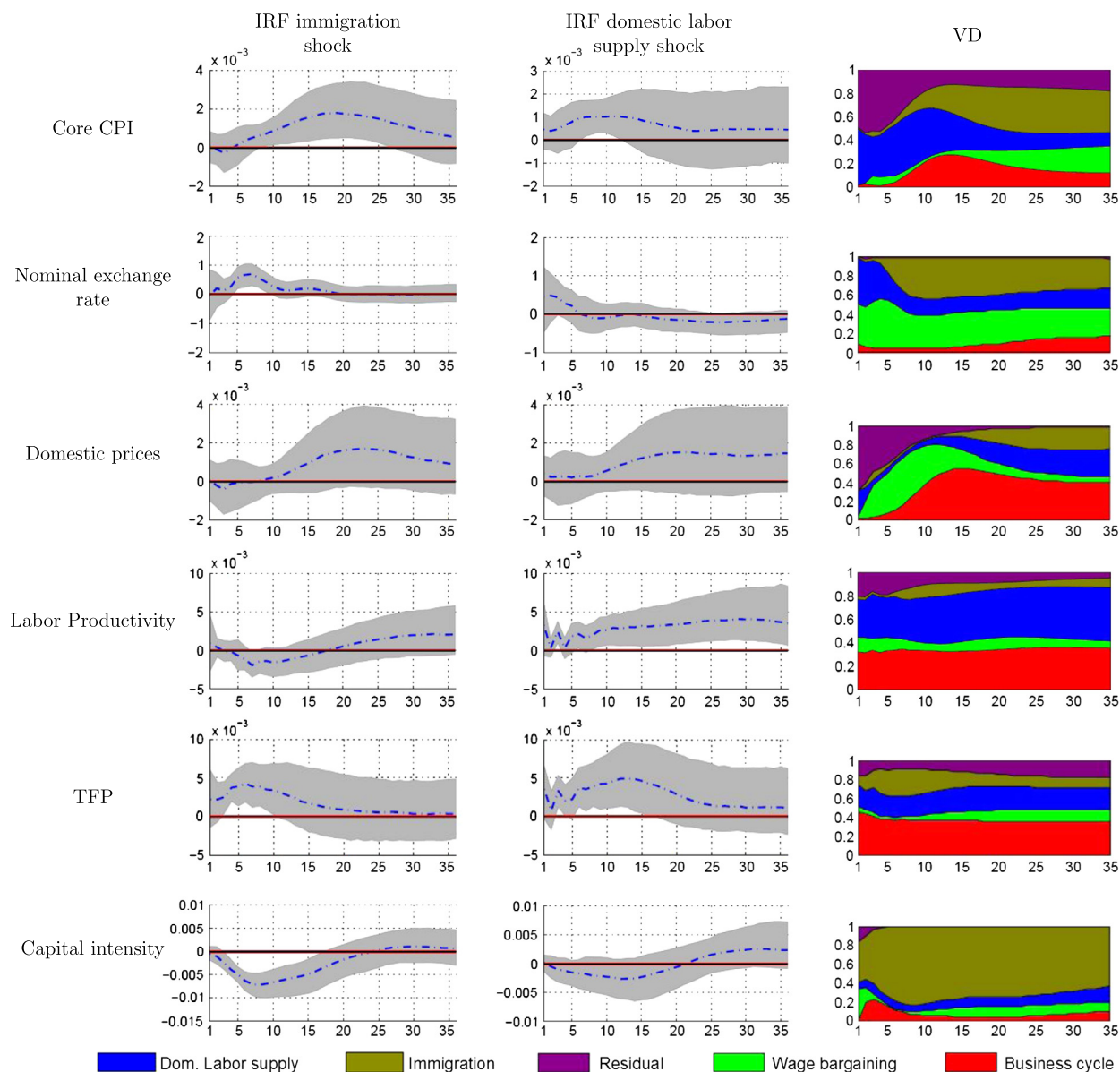


Fig. 11. Selected impulse responses and variance decomposition for some variables of interest.

impact (cf. first line on Fig. 10). Notice, however, that when we consider a measure of public spending in isolation, an exogenous increase in immigration leads to an impact decline in public spending followed by an increase in the medium run with a peak corresponding to the peak in the immigration response (cf. Fig. 3). This medium run increase in government spending may capture the impact of family reunifications, while the short run decline may reflect reduced expenses for unemployment benefits in response to the improved state of the labor market discussed in the previous section. Notably, the impulse responses for net fiscal revenues and government spending can be reconciled only with a positive effect on tax revenues in the medium run. Thus, our result confirms previous findings emerging from the microeconomic literature: the net fiscal impact of an exogenous increase in immigration is relatively small (even slightly positive in the short run) but the effects on both public spending and tax revenues are not negligible, possibly in keeping with the effects discussed in Borjas (1999) and Storesletten (2000). Results are different in the case of an increase in immigration driven endogenously by a negative domestic labor supply shock, which generates a negative conditional correlation between immigration and the state of public finances. In this case, however, it is reasonable to think that the worsening in public finances is driven by the decline in participation from natives, and immigration may even mitigate the negative consequences of the shock.

Immigration and house prices. Immigration booms are often associated with housing booms. McDonald (2013) finds that net migration changes imply large positive effects on house prices in a SVAR identified with a recursive structure

on New Zealand data while D'Albis et al. (2017) find no significant effect on French data when considering non-European immigration. Other papers have shown that immigration has a positive impact on *average* house prices using disaggregate data from metropolitan areas (cf. Saiz, 2003).¹¹

As we can see in Fig. 10, according to our model, immigration shocks have no impact on house prices. Domestic labor supply shocks generate a negative conditional correlation between house prices and immigration but, as in the case of public finance, it is conceivable that the surge in immigration may mitigate the decline in house prices in response to a negative domestic labor supply shock. House prices respond strongly to the business cycle shock, and to some extent to domestic labor supply shocks whereas immigration shocks are almost irrelevant for house price dynamics, as it can be seen in the variance decomposition. More generally, we can conclude that immigration does not seem to play a big role in driving the recent housing boom in Norway. One possible explanation relates to the fact that a large share of immigrants is composed of Eastern European workers active in the construction sector. While many of these workers are unlikely to buy a house (at least in the short run), their contribution to the supply of new houses may be substantial.

Immigration and household credit growth. The connection between immigration and household credit has not been discussed in the literature to the best of our knowledge. Given the increasing attention devoted to household credit as an indicator of financial instability (together with house prices), we include it in one of our experiments. Note, however, that the stock of household debt is a largely predetermined variable, mainly representing credit accorded in the past and unresponsive to shocks almost by construction. In light of this observation, we use household credit in first differences as this is a good approximation for the new credit accorded in the period, a variable potentially more responsive to macroeconomic shocks. Nonetheless, we find that immigration shocks have no impact on household credit growth, unlike positive domestic labor supply shocks which have positive and persistent effects. The variable of interest is mainly driven by business cycle shocks and domestic labor supply shocks, whereas the role of immigration shocks and wage bargaining shocks is negligible, thus confirming our previous results for house prices. We conclude that the exogenous component of immigration does not pose clear challenges to financial stability, while the endogenous component of immigration (the one responding to the domestic labor supply factors) is negatively correlated with commonly used indicators of financial instability.

Immigration, prices and the exchange rate. The effect of an immigration shock on prices is not obvious. On the one hand, the wage mitigating effects of the increase in labor supply may put downward pressure on marginal costs and thus on prices, depending on the degree of price rigidity. On the other hand, the size (but also the composition) of aggregate demand for consumer goods changes with a larger population. If supply adjusts with a delay, we may expect an increase in prices. Lach (2007) explores the effects of the massive inflow of Russian Jewish immigrants into Israel during the 1990s and finds that prices of goods decreased. He attributes this result to the higher price elasticity and lower search costs of the new immigrants compared to the existing population. Cortes (2008) considers the effects on non-tradable goods and services (unlike Lach, 2007) in the US and finds a negative effect, in particular for low skill-intensive services. Frattini (2008) finds small effects on UK prices: immigration decreases the growth rate of prices for services and non-traded goods, whereas it tends to increase the prices of tradeable goods.

We introduce a measure of CPI prices as an unrestricted variable in our system and we find no effect on the impact of the shock (cf. Fig. 11). Nevertheless, CPI prices tend to increase in the medium run. Notably, this effect seems to be driven by a depreciation of the exchange rate, whereas a measure of domestic prices reacts substantially less to an immigration shock. While these effects are relatively small, in keeping with Frattini (2008), we uncover a new channel (the exchange rate channel) that, as far as we know, has not been discussed in the previous literature. Immigration shocks are important drivers of CPI prices in the medium run, whereas the sum of three labor market shocks captures a dominant share of fluctuations in the exchange rate. The effect on the exchange rate is quantitatively important as immigration shocks explain around 35% of exchange rate fluctuations at long horizons. In light of these results, investigating the link between labor supply and the exchange rate in theoretical models seems to be an interesting avenue for future research. One tentative interpretation is that the exchange rate depreciation may reflect the impact of remittances.

Immigration and productivity. The impact of immigration on productivity has been discussed in Peri (2012), who finds a strong positive association between immigration and total factor productivity (TFP) in US data. The main channel responsible for this result is task specialization. Peri and Sparber (2009) show that in states with large inflows of immigrants, natives with lower education tend to specialize in communication-intensive tasks, leaving more manual-intensive tasks to immigrants. The rebalancing produces task specialization based on comparative advantages and results in efficiency gains.

In our last experiment, we introduce labor productivity, measured as output per hour, as an unrestricted variable in our SVAR (cf. Fig. 11). We find that immigration shocks lower productivity in the medium run, thus confirming one of the conjectures proposed in the previous section to explain the short-lived effects of immigration on GDP. Notably, positive domestic labor supply shocks (associated with a decline in immigration) lead to a large positive effect on productivity. Therefore, both the exogenous and the endogenous component of immigration induce a decline in productivity. In fact, the variance decomposition suggests that the indirect effect (associated with the endogenous component) may be quantitatively more important to explain the possible negative effects of immigration on productivity.

¹¹ Notice that recent research has highlighted a negative impact of immigration on house prices *within* metropolitan areas in the US and in the UK due to the mobility response of the native population (cf. Saiz and Wachter, 2011, and Sá, 2014).

A negative conditional correlation between immigration and productivity is not surprising since the two series are negatively correlated over the sample period. However, this result seems in contrast with the evidence for the US proposed by Peri (2012). To further investigate this issue, we now try to disentangle the different components of labor productivity. As shown under general assumptions by Bosler et al. (2016), labor productivity can be decomposed into TFP, capital intensity and labor quality. We use data on a measure of TFP (that however does not disentangle labor quality) and on capital intensity to isolate the effects of immigration shocks on the different components of labor productivity. In a first experiment, we include our measure of TFP as an unrestricted variable in the SVAR. We see from Fig. 11 that a positive immigration shock has a significant positive effect on TFP. In contrast, when we include capital intensity, we find a strong negative effect and immigration shocks are the dominant drivers of capital intensity. These more granular results are now consistent with Peri (2012) who also find a positive effect on TFP, possibly driven by task specialization, and a negative (although not statistically significant) effect on capital intensity. The dominant effect of immigration shocks on capital intensity in Norway may capture the fact that immigration shocks induce the adoption of less capital intensive and more unskilled efficient technologies. In keeping with this view, Lewis (2011) finds that US manufacturing plants located in areas that experienced faster growth in immigration adopted significantly less machinery per unit of output.

5. Narrative sign restrictions

In this section, we extend our baseline model to include narrative information. Antolin-Diaz and Rubio-Ramirez (2018) developed a methodology which allows researchers to impose that around selected historical events structural shocks and/or historical decompositions agree with some narrative information. For example, it is possible to impose that around a quarter (or several quarters) one specific shock has to be positive (or negative) or that this shock has to be the main (or the least important) driver of a variable or more (less) important than all the other shocks combined for a specific variable. The first is a restriction on the sign of the structural shocks. The second and the third are both restrictions on the historical decompositions, named Type A and Type B respectively. Antolin-Diaz and Rubio-Ramirez (2018) focus their attention on oil and monetary shocks and show that imposing only a few narrative sign restrictions may sharpen and even change the inference of a SVAR originally identified via traditional sign restrictions.

We believe that our model with immigration shocks constitutes an appropriate framework to include some narrative sign restrictions. In fact, the opening of the Norwegian labor market to Eastern European citizens is associated with specific time periods where it is natural to assume that exogenous factors were the main drivers of immigration. We set a restriction over the period 2004:Q4 – 2005:Q3 and we implement it either by imposing that positive immigration shocks must be at play over the selected period or by imposing that the immigration shock must be the main driver of the immigration variable (Type A restriction). In fact, while the main EU enlargement happened in May 2004, the barriers to access the Norwegian labor market for Eastern European workers were removed only gradually. Therefore, we choose to implement the narrative restriction over the period 2004:Q4 – 2005:Q3 when Norway started experiencing a large increase in immigration, mainly from Poland and Lithuania. Notably, all results obtained in the baseline model are basically unaffected in both cases. Our baseline model was already driven by positive immigration shocks over the period 2004:Q4 – 2005:Q3 and the immigration shock was already the main driver of immigration, at least in most draws. This constitutes, we believe, a validation of our baseline set-up.

Interestingly, some of our results start changing only when we introduce either stricter restrictions or more numerous restrictions.

In a first extension, we implement the narrative restriction on immigration as a Type B restriction. Therefore, we require the immigration shock to be more important than all the other shocks combined for the immigration variable over the period 2004:Q4 – 2005:Q3. Our results are still by and large confirmed, as shown in Fig. 12, with one important exception: the variance decomposition for unemployment changes and the role of immigration shocks for unemployment dynamics is reduced (cf. Figs. 6 and 12).

In a second extension, we impose more numerous restrictions on the sign of shocks. In combination with the restriction on the sign of immigration shocks used above, we consider two additional restrictions on the sign of domestic labor supply and business cycle shocks at specific dates. In fact, our sample includes an important pension reform which raised significantly the participation of natives by increasing the retirement age. The details of the reform and its effects on the labor market are described in Galaasen (2017). Against this background, we impose that a positive domestic labor supply shock must be at play on the implementation date of the pension reform and on its aftermath, over the period 2011:Q1 – 2011:Q3. In addition, we impose also a restriction on the sign of business cycle shocks to capture the effect of the Great Recession in the US over the period 2008:Q1 – 2008:Q4. Once again, we remark from Fig. 13 that our main results are by and large confirmed also in this extension. Nevertheless, also in this case we find differences in the variance decomposition for unemployment. The immigration shock loses explanatory power and the residual shock, which most theoretical models (although not all) would classify as a productivity shock, becomes an important driver of the short-run variation in the unemployment rate.¹²

¹² The use of three narrative restrictions on the sign of the shocks reduces somewhat the model uncertainty around the pointwise median impulse responses. All impulse responses are presented in the Online Appendix. We conjecture that introducing more narrative restrictions would lead to a further reduction in the confidence bands, at the cost, however, of increasing the computational burden which is already large in a model with five shocks.

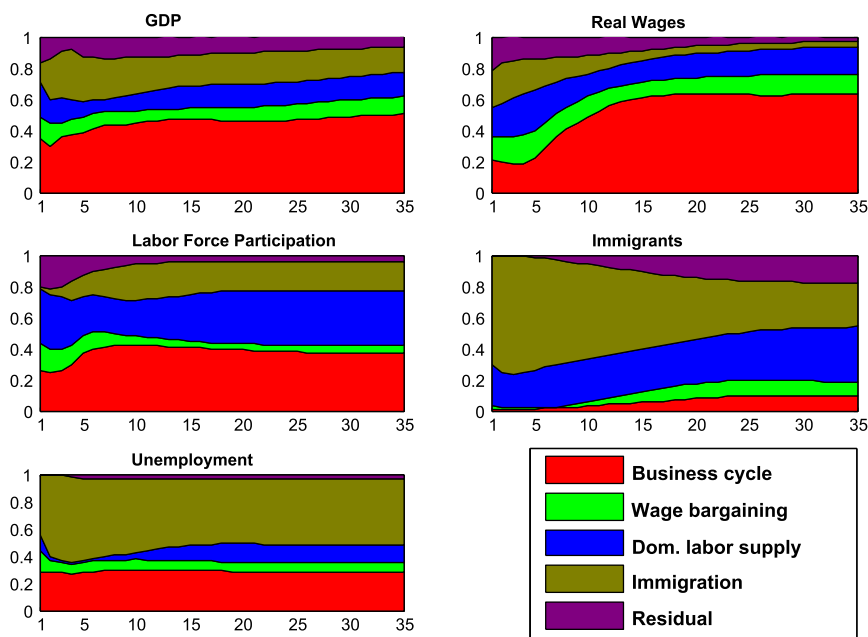


Fig. 12. Median forecast error variance decomposition at each horizon in the baseline model with narrative sign restrictions of type B on the immigration shocks in 2005.

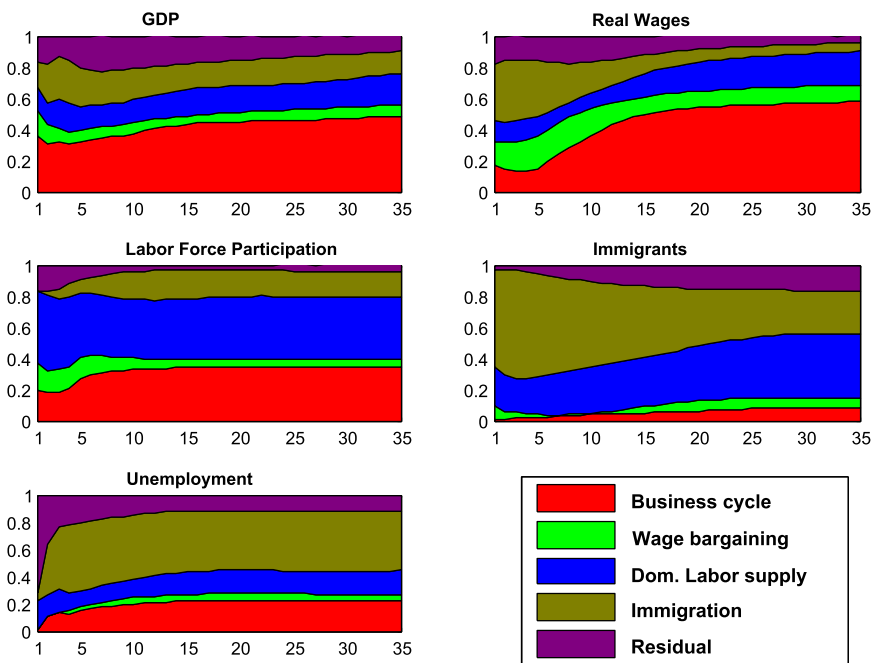


Fig. 13. Median forecast error variance decomposition at each horizon in the baseline with narrative restrictions on the sign of the immigration shocks in 2005, business cycle shocks in 2008 and domestic labor supply shocks in 2011.

We conclude that the introduction of narrative sign restrictions does not change the main results derived in the context of our baseline model. Nonetheless, these additional restrictions seem to play a role for unemployment dynamics by reducing the role of immigration shocks in the short-run, which was arguably large in our baseline model.

6. Conclusion

The economic impact of immigration is usually investigated in studies using detailed disaggregate data. This paper is one of the first attempts to include net immigration into the set of standard macroeconomic variables in a SVAR model

identified with a minimum set of sign restrictions. Our goal is to disentangle the drivers of immigration and the impact of immigration shocks on several variables that have been studied in the microeconomic literature. While a disaggregate approach can be more informative and detailed in several dimensions, an aggregate approach is needed to study business cycle fluctuations and possibly also to discuss some implications for macroeconomic policies.

We do not find any support for some of the arguments recently used against immigration in terms of native employment displacement effects and burden on public finances. On the contrary we find that an exogenous positive immigration shock in Norway lowers unemployment (even among native workers) and has a small positive effect on public finances in the short run. Moreover, an exogenous increase in immigration has no impact on house prices and household credit growth, a small positive effect on prices in the medium run and a negative effect on productivity. While the fears for employment and the balance of public finances seem misplaced, the negative impact on labor productivity may be worrisome for long-term growth.

We conclude by briefly discussing the implications of our analysis for financial stability and monetary policy. On the one hand, as already mentioned, the exogenous component of immigration does not seem to pose clear challenges to financial stability, while the endogenous component may even have a stabilizing effect. On the other hand, our SVAR framework is not equipped to discuss normative implications for monetary policy. Nevertheless, our analysis may give some indication of the effects of immigration on real economic activity and inflation, two key variables for monetary policy. In principle, the positive impact of immigration on the labor force may lead us to the conjecture that immigration has substantial effects on potential output. In practice, however, our analysis seems to downplay those effects. In fact, the negative effect on productivity limits the effects of the exogenous component of immigration on potential output, while the endogenous component of immigration reduces the effects of domestic labor supply shocks on potential output. The effects are larger if we consider the unemployment rate as a better indicator of real economic activity. As long as the natural rate of unemployment is relatively stable in response to exogenous variations in immigration, the fall in unemployment may indicate higher capacity utilization. The combination of a medium-run increase in inflation with higher capacity utilization suggests an expansionary effect of positive immigration shocks on key target variables for monetary policy.

Appendix A

A.1. Bayesian estimation of the VAR

We illustrate in this Appendix the econometric procedure we use for the estimation of the different VAR models presented in the paper.

Estimation procedure

The VAR model described in (1) can be rewritten in a compact way as:

$$\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{U}, \quad (2)$$

where $\mathbf{Y} = [y_1 \dots y_T]'$, $\mathbf{B} = [C_B \ B_1 \dots B_p]'$, $\mathbf{U} = [u_1 \dots u_T]'$, and

$$\mathbf{X} = \begin{bmatrix} 1 & y'_0 & \dots & y'_{-p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & y'_{T-1} & \dots & y'_{T-p} \end{bmatrix}.$$

Finally, for convenience, we rewrite (2) into its vectorized form:

$$\mathbf{y} = (\mathbf{I}_n \otimes \mathbf{X})\boldsymbol{\beta} + \mathbf{u}, \quad (3)$$

where $\mathbf{y} = \text{vec}(\mathbf{Y})$, $\boldsymbol{\beta} = \text{vec}(\mathbf{B})$, $\mathbf{u} = \text{vec}(\mathbf{U})$, and with $\text{vec}()$ denoting columnwise vectorization. The error term $\mathbf{u}$ follows a normal distribution with a zero mean and variance-covariance matrix $\Sigma \otimes \mathbf{I}_T$.

The likelihood function in $\mathbf{B}$ and Σ is defined as:

$$L(\mathbf{B}, \Sigma) \propto |\Sigma|^{-\frac{T}{2}} \exp \left\{ -\frac{1}{2}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})'(\Sigma^{-1} \otimes \mathbf{X}'\mathbf{X})(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) \right\} \exp \left\{ -\frac{1}{2}\text{tr}(\Sigma^{-1}\mathbf{S}) \right\},$$

where $\mathbf{S} = ((\mathbf{Y} - \mathbf{X}\hat{\mathbf{B}})'(\mathbf{Y} - \mathbf{X}\hat{\mathbf{B}}))$ and $\hat{\boldsymbol{\beta}} = \text{vec}(\hat{\mathbf{B}})$ with $\hat{\mathbf{B}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$. We specify diffuse priors so that the information in the likelihood is dominant and these priors lead to a Normal-Wishart posterior. In more detail, we use a diffuse prior for $\boldsymbol{\beta}$ and Σ that is proportional to $|\Sigma|^{-(n+1)/2}$. The posterior becomes:

$$p(\mathbf{B}, \Sigma | \mathbf{Y}, \mathbf{X}) \propto |\Sigma|^{-\frac{T+n+1}{2}} \exp \left\{ -\frac{1}{2}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})'(\Sigma^{-1} \otimes \mathbf{X}'\mathbf{X})(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) \right\} \exp \left\{ -\frac{1}{2}\text{tr}(\Sigma^{-1}\mathbf{S}) \right\}. \quad (4)$$

The posterior in (4) is the product of a normal distribution for $\boldsymbol{\beta}$ conditional on Σ and an inverted Wishart distribution for Σ (see, e.g. Kadiyala and Karlsson, 1997 for the proof). We then draw $\boldsymbol{\beta}$ conditional on Σ from

$$\beta|\Sigma, \mathbf{Y}, \mathbf{X} \sim N(\hat{\beta}, \Sigma \otimes (\mathbf{X}'\mathbf{X})^{-1})$$

and Σ from

$$\Sigma|\mathbf{Y}, \mathbf{X} \sim IW(S, \nu),$$

where $\nu = (T - n) * (p - 1)$ and N representing the normal distribution and IW the inverted Wishart distribution. To obtain identification via sign restrictions, we follow the procedure described in Rubio-Ramirez et al. (2010) and implemented in Furlanetto et al. (2019).

A.2. Data sources

This subsection lists the sources of the data series used in this paper. When the original data series is at a monthly frequency, we take quarterly averages of monthly data. All data series enter the VARs in logs except for unemployment which enters in percent of the workforce.

Immigration rate: Stock of immigrants in the country in percent of population aged 15–74. We include immigrants from EU/EFTA countries, North America, Australia, New Zealand and Eastern Europe. Quarterly gross immigration is collected from Statistics Norway. Net immigration is calculated using interpolated yearly data for emigration. The stock of immigrants is accumulated net immigration. *Sources: Statistics Norway and Norges Bank*

Real wage: Seasonally adjusted wage costs per hour deflated with the price level (CPI-ATE). *Sources: Statistics Norway and Norges Bank*

Participation rate: Seasonally adjusted workforce aged 15–74 in percent of population aged 15–74 from the Labor force survey. *Sources: Statistics Norway and Norges Bank*

Unemployment rate: Seasonally adjusted registered unemployment rate from the Norwegian Labour and Welfare Administration (NAV). *Sources: Statistics Norway and NAV*

GDP mainland Norway: Seasonally adjusted GDP mainland Norway (volumes) from national accounts. *Source: Statistics Norway*

Prices: Seasonally adjusted consumer price index adjusted for tax changes and excluding energy products (CPI-ATE). *Sources: Statistics Norway and Norges Bank*

Domestic Prices: Seasonally adjusted consumer price index domestic sources adjusted for tax changes and excluding energy products. *Sources: Statistics Norway and Norges Bank*

Exchange rate: Trade-weighted nominal exchange rate index (1–44) for 44 trading partners. *Sources: Thomson Reuters, Ecwin and Norges Bank*

Hours worked: Total hours worked in mainland Norway from national accounts adjusted for population growth. *Source: Statistics Norway*

Population: Population from 15 to 74. *Source: Statistics Norway*

House prices: Seasonally adjusted nominal house prices deflated by the CPI-ATE. *Sources: Statistics Norway, Eien-
domsmeglerforetakenes forening (EFF), Finn.no, Eiendomsverdi and Norges Bank*

Household credit: Seasonally adjusted C2 for households chained and break-adjusted deflated by the CPI-ATE and adjusted for population growth. Included in first differences. *Sources: Statistics Norway and Norges Bank*

Labor Productivity: Seasonally adjusted GDP mainland Norway (volumes) divided by hours worked. *Source: Statistics Norway*

TFP and capital intensity: Labor productivity decomposed into total factor productivity and capital intensity using a Cobb-Douglas production function. Data on capital, GDP, hours and labor income share for mainland Norway are used in the calculation. *Source: Statistics Norway and Norges Bank*

Government spending: Total expenditure from the quarterly central government fiscal account. Seasonally adjusted. *Sources: Statistics Norway and Norges Bank*

Public finances: Difference between tax income (excluding oil taxes) and government spending from the quarterly central government fiscal account. Seasonally adjusted. *Sources: Statistics Norway and Norges Bank*

Appendix B. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.red.2019.02.006>.

References

- Antolin-Diaz, J., Rubio-Ramirez, J., 2018. Narrative sign restrictions for SVARs. *The American Economic Review* 108, 2802–2829.
- Armstrong, J., McDonald, C., 2016. Why the drivers of migration matter for the labour market. Reserve Bank of New Zealand Analytical Notes 2016/02.
- Aukrust, O., 1977. Inflation in the open economy: a Norwegian model. In: Krause, L.B., Salant, W.S. (Eds.), *World Wide Inflation. Theory and Recent Experience*. Brookings, Washington DC.
- Ball, L.M., Leigh, D., Loungani, P., 2013. Okun's Law: Fit at Fifty. NBER Working Paper Series 18668.
- Bandeira, G., Caballe, J., Vella, E., 2018. Should I Stay or Should I Go? Austerity, Unemployment and Migration. Bank of Spain Working Paper.
- Barnichon, R., Figura, A., 2015. Declining desire to work and downward trends in unemployment and participation. NBER Macroeconomics Annual 2015.
- Bentolila, S., Dolado, J.J., Jimeno, J.F., 2008. Does immigration affect the Phillips curve? Some evidence for Spain. *European Economic Review* 52, 1398–1423.
- Bjørnland, H., Thorsrud, L.A., 2016. Boom or gloom? Examining the Dutch disease in two-speed economies. *The Economic Journal* 126, 2219–2256.
- Borjas, G., 1999. Immigration and welfare magnets. *Journal of Labor Economics* 17, 607–637.
- Borjas, G., 2003. The labor demand curve is downward sloping: reexamining the impact of immigration on the labor market. *The Quarterly Journal of Economics* 118, 1335–1374.
- Bosler, C., Daly, M., Fernald, L., Hobijn, B., 2016. The Outlook for US Labor-Quality Growth. Federal Reserve Bank of San Francisco, Working Paper 2016-14.
- Bratsberg, B., Raaum, O., 2012. Immigration and wages: evidence from construction. *The Economic Journal* 122, 1177–1205.
- Bratsberg, B., Raaum, O., Roed, M., Schöne, P., 2014. Immigration wage effects by origin. *Scandinavian Journal of Economics* 116, 356–393.
- Braun, S., Weber, H., 2016. How do Regional Labor Markets Adjust to Immigration? A Dynamic Analysis for Post-War Analysis. Deutsche Bundesbank Discussion Paper 05/2016.
- Campolmi, A., Gnocchi, S., 2016. Labor market participation, unemployment and monetary policy. *Journal of Monetary Economics* 79, 17–29.
- Canova, F., Ravn, M.O., 1998. Crossing the Rio Grande: Migrations and the Welfare State. CEPR Working Paper 2040.
- Canova, F., De Nicoló, G., 2002. Monetary disturbances matter for business cycle fluctuations in the G7. *Journal of Monetary Economics* 49, 1131–1159.
- Canova, F., Paustian, M., 2011. Business cycle measurement with some theory. *Journal of Monetary Economics* 58, 345–361.
- Card, D., 2005. Is the new immigration really so bad? *The Economic Journal* 115, 300–323.
- Chang, Y., Schorfheide, F., 2003. Labor-supply shifts and economic fluctuations. *Journal of Monetary Economics* 50, 1751–1768.
- Christiano, L., Eichenbaum, M., Trabandt, M., 2015. Understanding the Great Recession. *American Economic Journal: Macroeconomics* 7, 110–167.
- Cortes, P., 2008. The effect of low-skilled immigration on US prices: evidence from CPI data. *Journal of Political Economy* 116, 381–422.
- D'Albis, H., Boubtane, E., Coulibaly, D., 2015. Immigration policy and macroeconomic performance in France. *Etudes Et Documents* 5. CERDI.
- D'Albis, H., Boubtane, E., Coulibaly, D., 2017. International Migration and Regional Housing Markets: Evidence from France. IZA Discussion Paper 10516.
- D'Albis, H., Boubtane, E., Coulibaly, D., 2018. Immigration and Government Spending in OECD Countries. Paris School of Economics Working Paper 2018-41.
- Faust, J., 1998. The robustness of identified VAR conclusions about money. *Carnegie-Rochester Conference Series on Public Policy* 49, 207–244.
- Fry, R., Pagan, A., 2011. Sign restrictions in structural vector autoregressions. A critical review. *Journal of Economic Literature* 49, 938–960.
- Foroni, C., Furlanetto, F., Lepetit, A., 2018. Labor supply factors and economic fluctuations. *International Economic Review* 59, 1491–1510.
- Frattini, T., 2008. Immigration and prices in the UK. Manuscript.
- Furlanetto, F., Ravazzolo, F., Sarferaz, S., 2019. Identification of financial factors in economic fluctuations. *The Economic Journal* 129, 311–337.
- Galaasen, S., 2017. Pension Reform Disabled. Norges Bank Working Paper 20/2017.
- Gali, J., Smets, F., Wouters, R., 2011. Unemployment in an estimated New Keynesian model. In: NBER Macroeconomics Annual 2011. University of Chicago Press, pp. 329–360.
- Grangård, H., Nordbø, E., 2012. High immigration to Norway: who comes and why do they come? Norges Bank Staff Memo 25/2012.
- Hagelund, K., Nordbø, E., Wulfsberg, F., 2011. The effects of immigration. Norges Bank Staff Memo 12/2011.
- Hauser, D., Seneca, M., 2018. Labor mobility in a monetary union. Manuscript.
- Kadiyala, R., Karlsson, S., 1997. Numerical methods for estimation and inference in Bayesian VAR models. *Journal of Applied Econometrics* 12, 99–132.
- Kerr, S.P., Kerr, W., 2011. Economic Impacts of Immigration: a Survey. NBER working paper 16736.
- Kiguchi, T., Mountford, A., 2017. Immigration and unemployment: a macroeconomic approach. *Macroeconomic Dynamics*, 1–27.
- Lach, S., 2007. Immigration and prices. *Journal of Political Economy* 115, 548–587.
- Lewis, E., 2011. Immigration, skill mix, and capital skill complementarity. *The Quarterly Journal of Economics* 126, 1029–1069.
- Lozej, M., 2018. Economic migration and business cycles in a small open economy. *Economic Modelling*. <https://doi.org/10.1016/j.econmod.2018.07.012>. Forthcoming.
- Mandelman, F.S., Zlate, A., 2012. Immigration, remittances and business cycles. *Journal of Monetary Economics* 59, 196–213.
- McDonald, C., 2013. Migration and the housing market. Reserve Bank of New Zealand Analytical Notes 2013/10.
- Ottaviano, G., Peri, G., 2008. Immigration and National Wages: Clarifying the Theory and the Empirics. NBER Working Paper 14188.
- Ottaviano, G., Peri, G., 2012. Rethinking the effect of immigration on wages. *Journal of the European Economic Association* 10, 152–197.
- Peersman, G., 2005. What caused the early millennium slowdown? Evidence based on vector autoregressions. *Journal of Applied Econometrics* 20, 185–207.
- Peri, G., 2012. The effect of immigration on productivity: evidence from U.S. states. *Review of Economics and Statistics* 94, 348–358.
- Peri, G., Sparber, C., 2009. Task specialization, immigration and wages. *American Economic Journal: Applied Economics* 1, 135–169.
- Rubio-Ramirez, J.F., Waggoner, D.F., Zha, T., 2010. Structural vector autoregressions: theory and identification algorithms for inference. *The Review of Economic Studies* 77, 665–696.
- Sá, F., 2014. Immigration and house prices in the UK. *The Economic Journal* 125, 1393–1424.
- Saiz, A., 2003. Room in the kitchen for the melting pot: immigration and rental prices. *Review of Economics and Statistics* 85, 502–521.
- Saiz, A., Wachter, S., 2011. Immigration and the neighborhood. *American Economic Journal: Economic Policy* 3, 169–188.
- Shapiro, M., Watson, M., 1988. Sources of business cycle fluctuations. NBER Macroeconomics Annual 3, 111–156.
- Sims, C., Stock, J., Watson, M., 1990. Inference in linear time series models with some unit roots. *Econometrica* 58, 113–144.
- Smith, C., Thoenissen, C., 2018. Migration and Business Cycle Dynamics. CAMA Working Paper 2018-20.
- Storesletten, K., 2000. Sustaining fiscal policy through immigration. *Journal of Political Economy* 108, 300–323.
- Uhlig, H., 2005. What are the effects of monetary policy on output? Results from an agnostic identification procedure. *Journal of Monetary Economics* 52, 381–419.
- Weiske, S., 2017. On the macroeconomic effects of immigration: A VAR analysis for the US. Manuscript.